

DETAILED DESCRIPTION OF THE INVENTION — MODULAR DEVICE ECOSYSTEMS AND INTERCHANGEABLE COMPONENT ARCHITECTURES

The intraluminal physiologic monitoring device may be implemented as part of a modular ecosystem in which individual components may be interchangeable, upgradeable, or configurable according to user needs, clinical requirements, or technological advancement.

In various embodiments, the device may include separable modules comprising sensing bodies, electronic cores, power modules, communication interfaces, retention assemblies, or outer protective shells. Modular construction may enable reuse of high-value electronic components while allowing disposable or replaceable sensing interfaces.

Interchangeable sensing modules may vary in geometry, sensor density, sensing modality, or measurement capability. Users or clinicians may select modules optimized for specific physiologic measurements, comfort preferences, or research applications.

Certain embodiments may employ standardized mechanical and electrical interfaces allowing rapid attachment or detachment of components. Interfaces may include magnetic couplings, sealed electrical contacts, mechanical locking structures, threaded interfaces, snap-fit assemblies, or compliant connector systems.

Firmware and computational configurations may automatically adapt to installed modules through device identification protocols, enabling dynamic reconfiguration of sensing algorithms, calibration routines, or data processing workflows.

Modular architectures may support upgrade pathways in which newer sensor technologies, improved batteries, or enhanced communication modules may be integrated without replacing the entire device platform. Such designs may extend product lifecycle while maintaining compatibility with existing computational ecosystems.

In certain implementations, modularity may enable multi-user deployment models including shared electronic cores paired with individualized hygienic sensing components. Clinical environments may benefit from replaceable sterile modules while maintaining centralized processing hardware.

The modular ecosystem concept further allows expansion into broader distributed sensing platforms comprising multiple interoperable devices coordinated through shared software environments or cloud-based systems.

The modular device architectures described herein are illustrative rather than limiting. Any configuration allowing interchangeable, upgradeable, or configurable intraluminal monitoring components capable of achieving substantially similar functional outcomes is considered within the scope of the present disclosure.